Vector space $\mathbb{R}^n$

Goals

Matrix involves numbers and vectors - columns of matrix. Instead of individual columns, we look at "spaces" of vector to understand deeper, e.g. about equation $A\mathbf{x} = \mathbf{b}$

1. Vector space $\mathbb{R}^n$
2. Subspaces
3. Spanning
4. Independence
5. Basis and Dimensions
6. Rank of a matrix
7. Represent solutions to a homogeneous system of linear equations as linear combinations of basis solutions

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   - Independence
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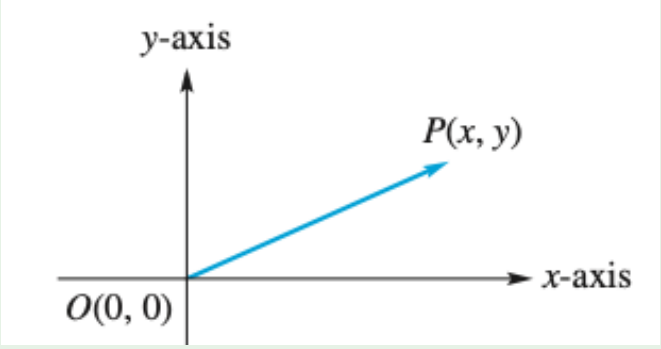
3. Rank of a matrix
4. Coordinates and Change Bases

Vector in $\mathbb{R}^n$

Example

- $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$: a vector in $\mathbb{R}^2$
- $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$: a vector in $\mathbb{R}^3$



Definition

Vector in $\mathbb{R}^n$ is a $n \times 1$ matrix $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$

A vector in $\mathbb{R}^n$ is also called a point in $\mathbb{R}^n$

In some textbooks, a vector in $\mathbb{R}^n$ is a $1 \times n$ matrix $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$.

- Vector zero $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$
- Two vectors are equal if and only if corresponding entries are the same

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \iff a_i = b_i \quad \forall i$$

Vector addition and scalar multiplication in $\mathbb{R}^n$

Let

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad and \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$$

be two vectors in $\mathbb{R}^n$

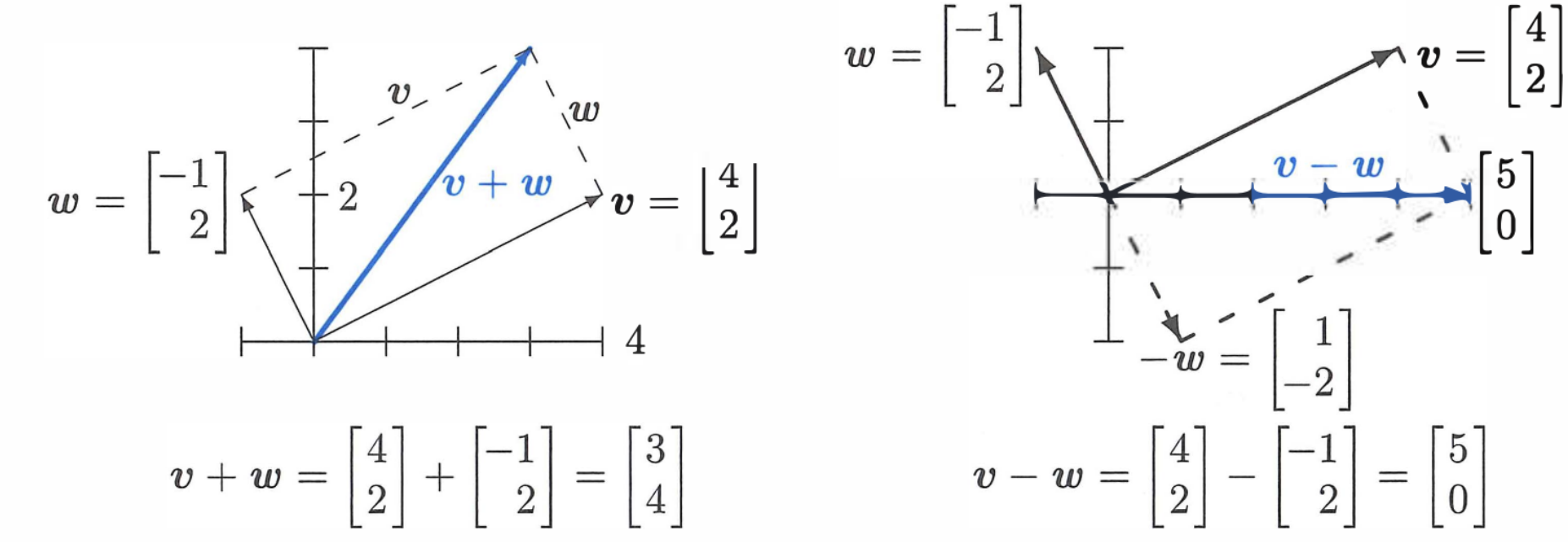
- Vector addition

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \dots \\ x_n + y_n \end{bmatrix}$$

- Scalar multiplication

$$a\mathbf{x} = \begin{bmatrix} ax_1 \\ ax_2 \\ \dots \\ ax_n \end{bmatrix}$$

Geometric meaning of vector addition



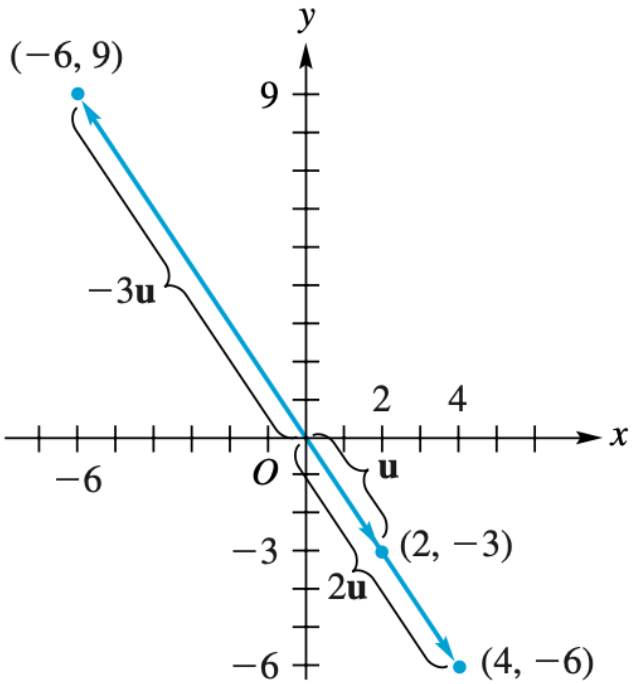
$$v + w = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$v - w = \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

Figure 1: Parallelogram rule for vector addition

Geometric meaning of scalar multiplication

If $k > 0$ then $k\mathbf{x}$ has the same direction as $\mathbf{x}$ but if $k < 0$ then $k\mathbf{x}$ has opposite direction.



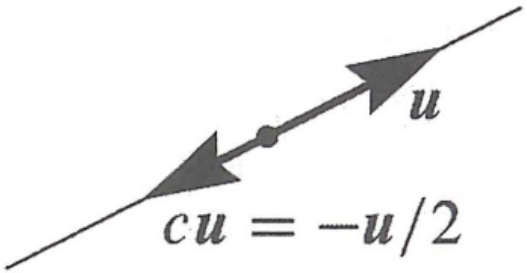


Figure 2: Line contain all scalar product of a vector

Set of all vectors generated by scalar product with a given vector is a line which is parallel with the direction of this vector

Example

Let $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$ then

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 5 \\ 0 \\ 3 \\ 1 \end{bmatrix} \qquad 3\mathbf{x} = \begin{bmatrix} 6 \\ -3 \\ 3 \\ 6 \end{bmatrix} \qquad 3\mathbf{x} + 2\mathbf{y} = \begin{bmatrix} 12 \\ -1 \\ 7 \\ 4 \end{bmatrix}$$

Vector space $(\mathbb{R}^n, +, .)$

Definition (Vector space $\mathbb{R}^n$)

Set of all n - vector with addition and scalar multiplication is a vector space, called vector space $\mathbb{R}^n$

Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and any numbers (scalars) a, b, we have

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
4. $\mathbf{u} + (\text{-}\mathbf{u}) = \mathbf{0}$
5. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
6. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
7. $a(b\mathbf{u}) = (ab)\mathbf{u}$
8. $1\mathbf{u} = \mathbf{u}$

Subspace of $\mathbb{R}^n$

A set U of vectors in $\mathbb{R}^n$ is called a **subspace** in $\mathbb{R}^n$ it satisfies the following properties:

1. If $\mathbf{u} \in U$ and $\mathbf{v} \in U$ then $\mathbf{u} + \mathbf{v} \in U$.
2. If $\mathbf{v} \in U$ then $a\mathbf{v} \in U$ for every real number a.

Subspace is a set of vectors which is closed under addition and scalar multiplication

Suspace always contains zero vector $\mathbf{0}$

Example

$U = \{\mathbf{0}\}$ consisting of only the zero vector is a subspace which is called **the zero subspace** because it satisfies all two conditions

- $\mathbf{0} + \mathbf{0} = \mathbf{0} \in U$
- For all a then $a\mathbf{0} = \mathbf{0} \in U$

Example

Consider a set of vector $S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\}$ in $\mathbb{R}^3$

Intuitively, we embed the vector space $\mathbb{R}^2$ into vector space $\mathbb{R}^3$. Then S is a subspace of $\mathbb{R}^3$ because it satisfies all 2 conditions

1. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ are in S then $\mathbf{x} + \mathbf{y}$ is also in S

   Why?

2. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is in S then so is $k\mathbf{x}$ for all number k

   Why?

Example

Plane M through the origin in $\mathbb{R}^3$

$$M = \{\mathbf{v} = (x, y, z)^T \in \mathbb{R}^3 : ax + by + cz = 0\}$$

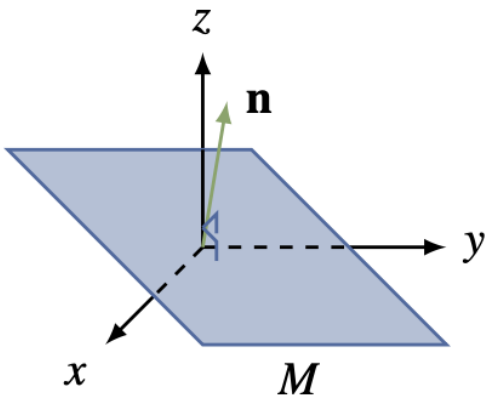
is a subspace of $\mathbb{R}^3$
It satisfies all 2 conditions

1. If both $\mathbf{v} = (v_1, v_2, v_3)^T$ and $\mathbf{u} = (u_1, u_2, u_3)^T$ are in M then $av_1 + bv_2 + cv_3 = 0$ and $au_1 + bu_2 + cu_3 = 0$. Hence

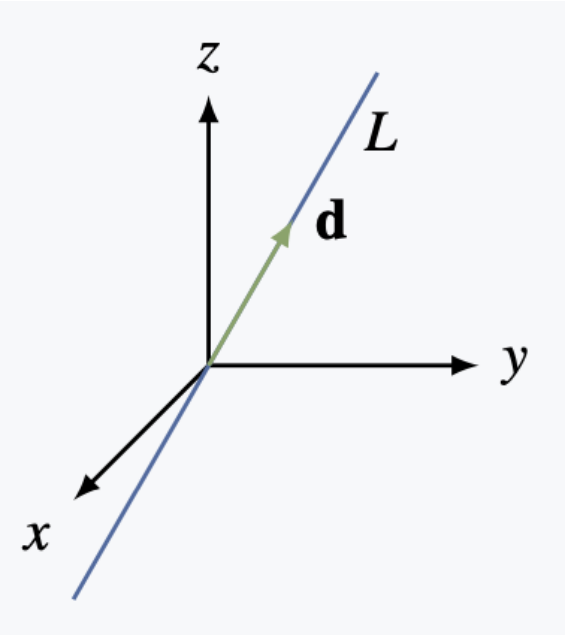
$$a(u_1 + v_1) + b(u_2 + v_2) + c(u_3 + v_3) = 0$$

   so $\mathbf{u} + \mathbf{v} \in M$

2. If $\mathbf{v} \in M$ then $kav_1 + kbv_2 + kcv_3 = 0$ for all k. Hence $k\mathbf{v} \in M$



Exercise



Prove that line through origin in $\mathbb{R}^3$

$$L = \{\mathbf{v} = (x, y, z)^T : x = at, y = bt, z = ct \text{ for some } t \in \mathbb{R}\}$$

is a subspace of $\mathbb{R}^3$

Not subspace example

- $U = \{(x, y)^T : x > 0, y > 0\}$
- $V = \{(x, y, z)^T : x + y + z = 1\}$
- $W = \{(a + b - 1, a, b)^T \mid a, b \in \mathbb{R}\}$

Definition (Linear combination)

Given vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ in $\mathbb{R}^n$, a **linear combination** of $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ is a vector of form

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 \cdots + t_k\mathbf{x}_k$$

where the t_i are scalars, called by the **coefficient** of x_i in the linear combination

Example

$\mathbf{w} = \begin{bmatrix} 12 \\ -1 \\ 7 \\ 4 \end{bmatrix} = 3\mathbf{x} + 2\mathbf{y}$ where $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$. So $\mathbf{w}$ is a linear combination of $\mathbf{x}$ and $\mathbf{y}$

Write a vector as a linear combination of other vectors

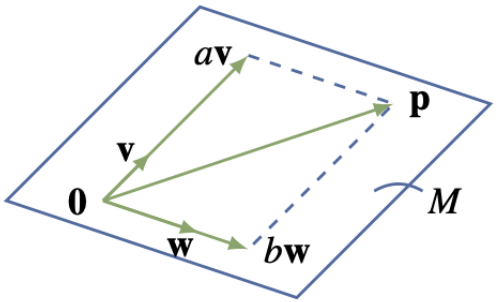
Given $\mathbf{u} = [1, -1, 2]^T, \mathbf{v} = [2, 1, 3]^T, \mathbf{w} = [0, -3, 1]^T$.
Write $\mathbf{w}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$, that is find numbers a, b such that

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}$$

$$(0, -3, 1)^T = a(1, -1, 2)^T + b(2, 1, 3)^T$$
$$(0, -3, 1)^T = (a + 2b, -a + b, 2a + 3b)^T$$
$$\rightarrow \begin{cases} a + 2b = 0 \\ -a + b = -3 \\ 2a + 3b = 1 \end{cases} \rightarrow \begin{cases} a = 2 \\ b = -1 \end{cases}$$

So

$$\mathbf{w} = 2\mathbf{u} - \mathbf{v}$$



A vector **p** is in M if and only if it has the form

$$\mathbf{p} = a\mathbf{v} + b\mathbf{w}$$

for certain real numbers a and b (**p** is a **linear combination** of **v** and **w**). So we can describe M as

$$M = \{a\mathbf{v} + b\mathbf{w} \ \mid \ a, b \in \mathbb{R}\}$$

We say $\{\mathbf{v}, \mathbf{w}\}$ is a *spanning set* of M

Given vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ in $\mathbb{R}^n$, a **linear combination** of $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ is a vector of form

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 \cdots + t_k\mathbf{x}_k$$

where the t_i are scalars

Definition (Span)

The **set of all linear combinations** of $\mathbf{x}_i$ is called the **span** of the $\mathbf{x}_i$ and is denoted

$$span(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k) = \{t_1\mathbf{x}_1 + t_2\mathbf{x}_2 \cdots + t_k\mathbf{x}_k \ \mid \ t_i \in \mathbb{R}\}$$

If $V = span(\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_k)$, we say that U is spanned by the vectors $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$, and that the vectors $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ span the space U.

Example

1. $span(\mathbf{x}) = \{t\mathbf{x} \mid t \in \mathbb{R}\}$
2. $span(\mathbf{x}, \mathbf{y}) = \{a\mathbf{x} + b\mathbf{y} \mid a, b \in \mathbb{R}\}$

Example

Let $\mathbf{x} = (2, -1, 2, 1)^T$ and $\mathbf{y} = (3, 4, -1, 1)^T$. Determine whether $\mathbf{p} = (0, -11, 8, 1)^T$ or $\mathbf{q} = (2, 3, 1, 2)^T$ are in $U = span(\mathbf{x}, \mathbf{y})$.

Solution

$\boldsymbol{p} \in U$ if and only if there exists a, b such that

$$\boldsymbol{p} = a\boldsymbol{x} + b\boldsymbol{y}$$

Equating components gives equations

$$2a + 3b = 0, \quad -a + 4b = -11, \quad 2a - b = 8, \quad a + b = 1$$

This system of linear equations has the solution $a = 3, b = -2$ so $\boldsymbol{p} \in U$
On the other hand, asking that $\boldsymbol{q} = a\boldsymbol{x} + b\boldsymbol{y}$ leads to equations

$$2a + 3b = 2, \quad -a + 4b = 3, \quad 2a - b = 1, \quad a + b = 2$$

which has no solution. So $\boldsymbol{q} \notin U$

Exercise

Find m such that $(-1, -2, m)^T$ lies in the subspace spanned by the vectors $(1, 2, -3)^T$, $(-1, -1, 5)^T$ and $(2, 5, -4)^T$.

Span theorem

$U = span(\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_k)$ in the vector space V

1. U is a subspace of V
2. If W is a subspace of V and $x_i \in W$ for all i then $U \subset W$

Example

If $\mathbf{x}$ and $\mathbf{y}$ are in $\mathbb{R}^n$, show that $span(\mathbf{x}, \mathbf{y}) = span(\mathbf{x} + \mathbf{y}, \mathbf{x} - \mathbf{y})$.

Solution

Denote $U = span(\boldsymbol{x}, \boldsymbol{y})$ and $W = span(\boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y})$. To prove that two subspaces $U = W$, we'll show that $U \subset W$ and $W \subset U$

- *Prove $W \subset U$ by showing $\boldsymbol{x} + \boldsymbol{y} \in U$ and $\boldsymbol{x} - \boldsymbol{y} \in U$ Since both $\boldsymbol{x} + \boldsymbol{y}$ and $\boldsymbol{x} - \boldsymbol{y}$ are linear combinations of **x** and **y**, we have both **x**+ **y** and **x**-**y** are the spanning set of **x** and **y**, that is **x**+ **y** and **x**-**y** are in U . Hence $W \subset U$*
- *Prove $U \subset W$ by showing $\boldsymbol{x} \in U$ and $\boldsymbol{y} \in U$: We have $\boldsymbol{x} = 0.5(\boldsymbol{x} + \boldsymbol{y}) + 0.5(\boldsymbol{x} - \boldsymbol{y})$ and $\boldsymbol{y} = 0.5(\boldsymbol{x} + \boldsymbol{y}) - 0.5(\boldsymbol{x} - \boldsymbol{y})$ are both linear combination of **x**+ **y** and **x**-**y**, so **x** and **y** belong to the spanning set of **x**-**y** and **x**-**y**, both **x** and **y** are in W. Hence $U \subset W$.*

Exercise

Prove that for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ then

$$span(\mathbf{x}, \mathbf{y}) \subset span(\mathbf{x}, \mathbf{y}, \mathbf{z})$$

Exercise

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Prove that $span(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = span(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$

Null Space as Spaning of Basic Solution

Given a matrix A, let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ be the basic solution to the system $A\mathbf{x} = \mathbf{0}$ obtained by the gaussian algorithm. We know that all the solution to $A\mathbf{x} = \mathbf{0}$ can be represented as a linear combination of $\mathbf{x}_1, \ldots, \mathbf{x}_k$, that is

$$\mathbf{x} = t_1\mathbf{x}_1 + \ldots t_k\mathbf{x}_k$$

for some scalar t_i. So

$$\text{null}A \subset span(\mathbf{x}_1, \ldots, \mathbf{x}_k)$$

Inversely, it is evident that $\mathbf{x}_i \in \text{null}A$ for all i. So

$$span(\mathbf{x}_1, \ldots, \mathbf{x}_k \subset \text{null}A$$

Hence $\text{null}A = span(\mathbf{x}_1, \ldots, \mathbf{x}_k)$

Image Space as Spanning of Column Vector

Let $\mathbf{c}_1, \ldots, \mathbf{c}_n$ denote the columns of the $m \times n$ matrix A. We have

$$\begin{bmatrix} A\mathbf{e}_1 & \ldots & A\mathbf{e}_n \end{bmatrix} = A \begin{bmatrix} \mathbf{e}_1 & \ldots & A\mathbf{e}_n \end{bmatrix} = A\mathbf{I}_n = A = \begin{bmatrix} \mathbf{c}_1 & \ldots & \mathbf{c}_n \end{bmatrix}$$

where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is the standard basis of $\mathbb{R}^n$.
So $A\mathbf{e}_i = \mathbf{c}_i$ for all i. That is $\mathbf{c}_i \in \text{im}A$ for all i and then

$$span(\mathbf{c}_1, \ldots, \mathbf{c}_n) \subset \text{im}A$$

Conversely, if $\mathbf{y} \in \text{im}A$ then there exists $\mathbf{x} = \begin{bmatrix} x_1 & \ldots & x_n \end{bmatrix}^T$ such that $A\mathbf{x} = \mathbf{y}$. In other words,

$$\mathbf{y} = A\mathbf{x} = x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n \text{ which is in } (\mathbf{c}_1, \ldots, \mathbf{c}_n)$$

So

$$\text{im}A \subset span(\mathbf{c}_1, \ldots, \mathbf{c}_n)$$

Hence

$$\text{im}A = span(\mathbf{c}_1, \ldots, \mathbf{c}_n)$$

- Spanning sets S provide vectors so that any vector in the space can be expressed as a linear combination of the members of S. We remarked that a vector space can have many different spanning sets and that spanning sets for the same space need not have the same number of vectors.
- How to determine a spanning set for a vector space $\mathbb{R}^n$ that is minimal, in the sense that it contains the fewest number of vectors, then we have an efficient way to describe every vector in $\mathbb{R}^n$

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If $U = span(\mathbf{x}_1, \mathbf{x}_2, ..., \ \mathbf{x}_k)$is a subspace of the vector space $(V, +, .)$, then every vector in U can be written as a linear combination of the $\mathbf{x}_i$ in at least one way. Our interest here is in spanning sets where each vector in U has an *exactly one representation* as a linear combination of these vectors.

Definition (Independence)

A set of vectors $\mathbf{x}_1, \mathbf{x}_2, ..., \ \mathbf{x}_k$ is called **linearly independent** (or simply **independent**) if it satisfies the following condition:

$$\text{If } t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + ... + t_k \ \mathbf{x}_k = 0 \text{ then } t_1 = t_2 = \cdots = t_k = 0$$

Theorem

If $\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \ \boldsymbol{x}_k$ is an independent set of vectors in a vector space V, then every vector in $span(\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \ \boldsymbol{x}_k)$ has a unique representation as a linear combination of the $\boldsymbol{x}_i$.

Independence Test

A set of vectors is independent if and only if the only linear combination that vanishes is the trivial one. Hence to verify that a set $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_k$ of vectors in $\mathbb{R}^n$ is independent, proceed as follows:

1. Set a linear combination equal to zero:

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + ... + t_k\ \mathbf{x}_k = 0$$

2. Show that $t_i = 0$ for each i (that is, the linear combination is trivial). Of course, if some nontrivial linear combination vanishes, the vectors are not independent

.

Example

Determine whether $\{(1,0,-2,5)^T, (2,1,0,-1)^T, (1,1,2,1)^T\}$ is independent in $\mathbb{R}^4$.

Solution

Suppose a linear combination vanishes:

$$r(1,0,-2,5)^T + s(2,1,0,-1)^T + t(1,1,2,1)^T = (0,0,0,0)^T$$

Equating corresponding entries gives a system of four equations:

$$r + 2s + t = 0, \quad s + t = 0, \quad -2r + 2t = 0, \quad 5r - s + t = 0$$

The only solution is the trivial one $r = s = t = 0$ (verify), so these vectors are independent

Example

Show that if $\{\mathbf{x}, \mathbf{y}\}$ is independent then $\{2\mathbf{x} + 3\mathbf{y}, \mathbf{x} - 5\mathbf{y}\}$ is independent

Example

Show that if $\{\mathbf{x}, \mathbf{y}\}$ is independent then $\{2\mathbf{x} + 3\mathbf{y}, \mathbf{x} - 5\mathbf{y}\}$ is independent

Solution

Set

$$a(2\boldsymbol{x} + 3\boldsymbol{y}) + b(\boldsymbol{x} - 5\boldsymbol{y}) = \boldsymbol{0} \tag{1}$$

which is equivalent to

$$(2a + b)\boldsymbol{x} + (3a - 5b)\boldsymbol{y} = \boldsymbol{0} \tag{2}$$

Since $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are independent, (2) happens if and only if*

$$\begin{cases} 2a + b = 0 \\ 3a - 5b = 0 \end{cases} \Rightarrow a = b = 0$$

So the set $\{2\boldsymbol{x} + 3\boldsymbol{y}, \boldsymbol{x} - 5\boldsymbol{y}\}$ *is independent.*

Example

The set of vectors

$$\mathbf{e}_1 = (1, 0, \ldots, 0)^T, \mathbf{e}_2 = (0, 1, \ldots, 0)^T, \ldots, \mathbf{e}_n = (0, 0, \ldots, 1)^T$$

is called the **standard basis** of $\mathbb{R}^n$. The standard basic is independent.

Exercise

1. Show that any set containing zero vector $\mathbf{0}$ is not independent.
2. Given a vector $\mathbf{x}$ then the set $\{\mathbf{x}\}$ is independent if and only if $\mathbf{x} \neq \mathbf{0}$

Theorem (Independent Test with Determinant)

A set of n vector $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ in $\mathbb{R}^n$ is linear independent if and only if $\det(A) \neq 0$ where A is the matrix whose columns (or rows) are the elements in S.

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A set of n vector $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ is linear independent if and only if $\det(A) \neq 0$ where A is the matrix whose columns (or rows) are the elements in S.

Proof.

Equation

$$x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n = 0$$

can be written in matrix form $A\mathbf{x} = \mathbf{b}$ has unique trial solution if and only if $\det(A) \neq 0$. $\square$

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can be written in matrix form $A\mathbf{x} = \mathbf{b}$ has unique trial solution if and only if $\det(A) \neq 0$. □

Corollary

If $\det(A) \neq 0$ then the columns of A span $\mathbb{R}^n$

Theorem (Independent Test with Inverse)

The following are equivalent for an $n \times n$ matrix A

1. A is invertible
2. The columns of A are linearly independent
3. The row of A are linearly independent

Example

Is $S = \left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T, \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}^T, \begin{bmatrix} 3 & 0 & -1 \end{bmatrix}^T \right\}$ linearly independent?

Solution

We form a matrix A whose columns are the vectors in S

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 0 \\ 3 & 2 & -1 \end{bmatrix}$$

Since $\det(A) = 2 \neq 0$, the colums of A are linearly independent. That is S is linearly independent.

Exercise

Which of the following sets are independent?

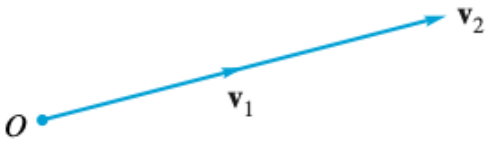
1. $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} \right\}$ in $\mathbb{R}^3$
2. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ in $\mathbb{R}^3$

Geometry Meaning of independence

- Two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^2$ are independent then there exists a_1 and a_2 not both zero such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 = 0$$

If $a_1 \neq 0$ then $\mathbf{v}_1 = -\frac{a_2}{a_1}\mathbf{v}_2$. If $a_2 \neq 0$ then $\mathbf{v}_2 = -\frac{a_1}{a_2}\mathbf{v}_1$



(a) Linearly dependent vectors in R^2.

(b) Linearly independent vectors in R^2.

- Three vectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ in $\mathbb{R}^3$ are independent then there exists a_1, a_2 and a_3 not both zero such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 = 0$$

say $a_2 \neq 0$ then $\mathbf{v}_2 = -\frac{a_1}{a_2}\mathbf{v}_1 - \frac{a_3}{a_2}\mathbf{v}_3$, that is $\mathbf{v}_2$ is in the subspace W spanned by $\mathbf{v}_1$ and $\mathbf{v}_3$.

(a) Linearly dependent vectors in R^3. (b) Linearly independent vectors in R^3.

Theorem

The nonzero vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_n$ *in a vector space* $\mathbb{R}^n$ *are linearly dependent if and only if one of the vectors* $\boldsymbol{v}_j, \quad (j \geq 2)$ *is a linear combination of the preceding vectors* $\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_{j-1}$.

Fundamental Theorem

Theorem

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$ and if $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is a set of $k \geq 1$ linearly independent vectors in U, then $k \leq m$.

Fundamental Theorem

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Proof.

Because $\mathbf{v}_1 \in U = span(\mathbf{u}_1, \ldots, \mathbf{u}_m)$, there exists numbers $a_1, \ldots, a_m$ such that

$$\mathbf{v}_1 = a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m$$

Since the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is independent, $\mathbf{v}_1 \neq \mathbf{0}$, not all a_i are zero, say $a_1 \neq 0$ (after relabelling the $\mathbf{u}_i$) then $\mathbf{u}_1 = \frac{1}{a_1}\mathbf{u}_1 - \frac{a_2}{a_1}\mathbf{u}_2 - \cdots - \frac{a_m}{a_1}\mathbf{u}_m$. So

$$U = span(\mathbf{v}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m)$$

□

continue proof.

Hence, write

$$\mathbf{v}_2 = b_1\mathbf{v}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{v}_m$$

If all c_i is zero then $\mathbf{v}_2 = b_1\mathbf{v}_2$ hence $\{\mathbf{v}_1, \mathbf{v}_2\}$ is dependent. It is contrary to the independent of $\mathbf{v}_i$. So $c_i \neq 0$ for some i, say $c_2 \neq 0$ (after possible relabelling of the $\mathbf{u}_i$. So

$$U = span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{u}_m)$$

If $k > m$ then this procedure continues until all the vector $\mathbf{u}_i$ are replaced by $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$. So

$$U = span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m)$$

Since $\mathbf{v}_{m+1} \in U$, $\mathbf{v}_{m+1}$ must be a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ which is contrary to the independence of $\mathbf{v}_i$.
Hence $k > m$ cannot be valid and so $k \leq m$

$\square$

Definition (Basis)

If U is a subspace of a vector space $\mathbb{R}^n$, a set $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\}$ of vectors in U is called a basis of U if it satisfies the following two conditions:

1. $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\}$ is linearly independent.
2. $U = span(\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\})$

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2. $U = span(\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\})$

Theorem (Invariance Theorem)

If $\{\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \boldsymbol{x}_m\}$ and $\{\boldsymbol{y}_1, \boldsymbol{y}_2, ..., \boldsymbol{y}_k\}$ are bases of a subspace U of a vector space $\mathbb{R}^n$, then $m = k$.

Proof of Invariance Theorem

We have U is spanned by m vectors $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m)$ and U contains the independent set of k vectors $\{\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_k\}$. By Fundamental theorem, we have $m \leq k$.
Similarly U is spanned by k vectors $\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_k)$ and U contains the independent set of m vectors $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\}$. By Fundamental theorem, we have $m \leq k$.
So $m = k$

Definition (Dimension of a Subspace of $\mathbb{R}^n$)

If U is a subspace of $\mathbb{R}^n$, a set $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_m\}$ is any basis of U then the number, m, of vectors in the basis is called the dimension of U , denoted

$$dim(U) = m$$

The dimension of the trivial vector space $\{\mathbf{0}\}$ is zero.

Example

$dim(\mathbb{R}^n) = n$ and $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n\}$ is a basis.

Example

$dim(\mathbb{R}^n) = n$ and $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n\}$ is a basis.
Every basis of $\mathbb{R}^n$ consists of n independent vectors.

Exercise

Which of the following is a basis of $\mathbb{R}^3$

1. $\{(4,2,0)^T, (1,4,1)^T, (1,-3,1)^T\}$
2. $\{(1,2,4)^T, (0,1,3)^T\}$
3. $\{(1,1,1)^T, (2,3,0)^T, (4,3,5)^T, (0,0,6)^T\}$

Example

Let $U = \left\{ \begin{bmatrix} r \\ s \\ r \end{bmatrix} | r, s \in \mathbb{R} \right\}$.

Show that U is a subspace in $\mathbb{R}^3$, find a basis and calculate $dim(U)$.

Example

Let $U = \left\{ \begin{bmatrix} r \\ s \\ r \end{bmatrix} | r, s \in \mathbb{R} \right\}$.

Show that U is a subspace in $\mathbb{R}^3$, find a basis and calculate $dim(U)$.

Solution

It is clear that $\begin{bmatrix} r \\ s \\ r \end{bmatrix} = r\boldsymbol{u} + s\boldsymbol{v}$ *where* $\boldsymbol{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ *and* $\boldsymbol{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. *So*

$U = span(\boldsymbol{u}, \boldsymbol{v})$ *which is a subspace in* $\mathbb{R}^3$.

We next show that $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *are independent. Hence* $\{\boldsymbol{u}, \boldsymbol{v}\}$ *is a basis of* U *and thus* $dim(U) = 2$.

In fact, $r\boldsymbol{u} + s\boldsymbol{v} = 0$ *if and only if* $\begin{bmatrix} r \\ s \\ r \end{bmatrix} = \boldsymbol{0}$ *so* $r = s = 0$.

Exercise

Find a basis and determine the dimension of the subspace

$$U = \left\{ \begin{bmatrix} a \\ a+b \\ a-b \\ b \end{bmatrix} : \quad a, b \in \mathbb{R} \right\}$$

in $\mathbb{R}^4$

Theorem

Let $U \neq \{\mathbf{0}\}$ be a subspace of $\mathbb{R}^n$. Then

1. *U has a basis and $dim(U) \leq n$*
2. *Any independent set in U can be enlarged (by adding vectors from the standard basis) to a basis of U*
3. *Any spanning set for U can be cut down (by deleting vectors) to a basis of U*

Theorem

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Proof.

1. Suppose that $dim(U) = m$. So a basic of U is has m vectors which is an independent set in $\mathbb{R}^n$.
   We have $\mathbb{R}^n$ is spanned by n vectors in the standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$.
   By Fundamental Theorem, $m \leq n$

$\square$

continue proof.

2. Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be an independent set in U. If $\mathbf{e}_1 \notin span\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ then $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{e}_1\}$ is an independent set. Inversely if $\mathbf{e}_1 \in span\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ then we do not add $\mathbf{e}_1$ to enlarge the current independent set $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. We continue this process to reach a basis of $\mathbb{R}^n$
3. If $U = \{\mathbf{0}\}$, it is clear.
   Suppose that $U = span(\mathbf{u}_1, \ldots, \mathbf{u}_m) \neq \{\mathbf{0}\}$ then at least one $\mathbf{u}_i$ is non zero, namely $\mathbf{u}_1$ (after posssible relabeling the index). Then $\{\mathbf{u}_1\}$ is an independent set. Continue the process of enlarging an independent set in U by the vector in the spanning set as in the previous part, we obtain a basis of U.

$\square$

As consequence, we have

Theorem

Let U be a subspace of $\mathbb{R}^n$ where $dim(U) = m$ and $B = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$. Then B is independent if and only if B is a basis of U, in other words B spans U

Theorem

Let $U \subseteq V$ be subspaces in $\mathbb{R}^n$ then

1. $dim(U) \leq dim(V)$
2. *If $dim(U) = dim(V)$ then $U = V$*

Example

Find a basis of $\mathbb{R}^4$ containing $S = \{\mathbf{u}, \mathbf{v}\}$ where $\mathbf{u} = \begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}^T$, $\mathbf{v} = \begin{bmatrix} 2 & -1 & 0 & 1 \end{bmatrix}^T$.

Example

Find a basis of $\mathbb{R}^4$ containing $S = \{\mathbf{u}, \mathbf{v}\}$ where $\mathbf{u} = \begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}^T$, $\mathbf{v} = \begin{bmatrix} 2 & -1 & 0 & 1 \end{bmatrix}^T$.

Solution

We can find such a basis by adding vectors from the standard basis of $\mathbb{R}^4$ to S

- *try with $\boldsymbol{e}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T$, we see that $\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{e}_1\}$ is independent*
- *Now add another vector from the standard basis, say $\boldsymbol{e}_2$ and find that $B = \{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{e}_1, \boldsymbol{e}_2\}$ is independent.*
- *Since B has $4 = dim(\mathbb{R}^4)$ vectors, B is a basis of $\mathbb{R}^4$.*

Theorem (Spanning Set Theorem)

Let $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ *and* $U = span(S)$

1. *If one of the vector in* S *- say,* $\boldsymbol{v}_i$ *is a linear combination of the remaining vectors in* S *then the set formed from* S *by removing* $\boldsymbol{v}_k$ *still spans* U
2. *If* $H \neq \{\boldsymbol{0}\}$ *then some subset of* S *is a basis for* U

Example

Find a basis and the dimension of the following subspace

$$U = span\left\{\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\-1\\1\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$$

Solution

We enlarge an independent set from spanning set of U step by step to obtain a basis

1. $\boldsymbol{v}_1$ *is not a zero vector so* $\{\boldsymbol{v}_1\}$ *is an independent set*
2. *For any number* k, $\boldsymbol{v}_2 \neq k\boldsymbol{v}_1$. *Thus* $\boldsymbol{v}_2 \notin span\{\boldsymbol{v}_1\}$ *so* $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ *is independent*
3. $\boldsymbol{v}_3 = \frac{1}{2}\boldsymbol{v}_1 - \frac{1}{1}\boldsymbol{v}_2$ *that is* $\boldsymbol{v}_3 \in span\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$

Hence $U = span(\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3) = span(\boldsymbol{v}_1, \boldsymbol{v}_2)$ *So* $B = \{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ *is independent then* B *is a basis of* U *and* $dim(U) = 2$

Exercise

Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$

Solution space of homogeneous system of linear equations

General solution of the system of homogeneous linear equations

$$\begin{aligned} x_1 - 2x_2 + 3x_3 - 2x_4 &= 0 \\ -3x_1 + 6x_2 + x_3 &= 0 \\ -2x_1 + 4x_2 + 4x_3 - 2x_4 &= 0 \end{aligned}$$

is $x_1 = 2s + \frac{1}{5}t$, $x_2 = s, x_3 = \frac{3}{5}t, x_4 = t$ and can be written in matrix form

$$\mathbf{x} = \begin{bmatrix} 2s + \frac{1}{5}t \\ s \\ \frac{3}{5}t \quad t \end{bmatrix} = \begin{bmatrix} 2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{5}t \\ 0 \\ \frac{3}{5}t \\ t \end{bmatrix} = s\mathbf{x}_1 + t\mathbf{x}_2$$

where $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$ which are called **basis solutions**

Solution Space

Set of all the solutions of the system equation is $span(\mathbf{x}_1, \mathbf{x}_2)$ which is called by **solution space**

1. basis solution $\mathbf{x}_1, \mathbf{x}_2$ are independent
2. dimension of solution space is equal to number of solution in a basis solutions set

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Row space - Column space

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \vdots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

be a $m \times n$ matrix

- **Row space** of A, denoted by $row(A)$, is a subspace spanned by the rows of A
- **Column space** of A, denoted by $col(A)$, is a subspace spanned by the columns of A
- The dimension of the row (column) space of A is called the **row (column) rank** of A.

Lemma

Let A and B denote $m \times n$ matrices.

1. *If $A \to B$ by elementary row operations, then $row(A) = row(B)$.*
2. *If $A \to B$ by elementary column operations, then $col(A) = col(B)$.*

Theorem (Rank Theorem)

Let A be an $m \times n$ matrix then

$$dim(col(A)) = dim(row(A)) = rank(A)$$

If we can carry $A \to R$ by elementary row operations where R is a row-echelon matrix then

1. *The nonzero rows of R are a basis of $row(R)$.*
2. *The columns of R containing leading ones are a basis of $col(R)$.*

Proof of Rank Theorem

1. It is clear
2. We can write $R = UA$ where U is the product of elementary matrix in the reduction procedure and is invertible. Write $A = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n \end{bmatrix}$ are $\mathbf{c}_i$ are columns of A then

$$R = U \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n \end{bmatrix} = \begin{bmatrix} U\mathbf{c}_1 & U\mathbf{c}_2 & \dots & U\mathbf{c}_n \end{bmatrix}$$

   Let $\mathbf{c}_{j_1}, \dots, \mathbf{c}_{j_r}$ denote the columns of R containing leadling 1s. Then $\{\mathbf{c}_{j_1}, \dots, \mathbf{c}_{j_r}\}$ is independent because the leading 1s are in different rows and have zeros below and to the left of them. Other column in R can be written as a linear combination of $\mathbf{c}_{j_1}, \dots, \mathbf{c}_{j_r}$. Hence $\{\mathbf{c}_{j_1}, \dots, \mathbf{c}_{j_r}\}$ is a basis of $col(R) = col(A)$.

$dim(col(A))$ and $dim(row(A))$ are equal to the leading 1s and hence equal to $rank(A)$

Example

Compute the rank of $A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$ and find a basis of $row(A)$ and $col(A)$

Solution

The reduction of A to row-echelon form is

$$\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix} \xrightarrow[r_3-r1]{r_2-3r_1} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{bmatrix} \xrightarrow[r_3-r2]{-r_2} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- $rank(A) = 2$
- $\left\{ \begin{bmatrix} 1 & 2 & 2 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & -3 \end{bmatrix} \right\}$ *is a basis of* $row(A)$
- *The leading 1s are in columns 1 and 3* $\Rightarrow$ *column 1 and 3 of* A *are a basis of* $col(A)$*, that is* $\left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \right\}$

Exercise

Find a basis of column space, row space and rank of matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 1 & 1 \\ 4 & -2 & 3 \\ -6 & 3 & 0 \end{bmatrix}$$

Example

Find a basis and the dimension of the following subspace

$$U = span\left\{\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\-1\\1\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$$

Example

Find a basis and the dimension of the following subspace

$$U = span\left\{\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\-1\\1\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$$

Solution

U is $col(A)$ where

$$A = \begin{bmatrix}1 & 1 & 0\\1 & -1 & 1\\1 & 1 & 0\end{bmatrix}$$

Process as the previous example to find a basis of $col(A)$ and then of U

Exercise

Find a basis of the following subspace

1. $U = span\left(\begin{bmatrix} 1 & -1 & 0 & -3 \end{bmatrix}, \begin{bmatrix} 2 & 1 & 5 & 1 \end{bmatrix}, \begin{bmatrix} 4 & -2 & 5 & 7 \end{bmatrix}\right)$
2. $V = span\left(\begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -8 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -10 \end{bmatrix}\right)$

Rank of Matrix Properties

1. $rank(A) = rank(A^T)$
2. If A is an $m \times n$ matrix then $rank(A) \leq m$ and $rank(A) \leq n$
3. For any invertible matrices U and V with convenient sizes, we have $rank(A) = rank(UA) = rank(AV)$

Definition (Null space)

- The **null space** of a $m \times n$ matrix A is the solution space of $A\mathbf{x} = \mathbf{0}$
- **nullity** of A is the dimension of the null space of A

Definition (Image space)

The **image space** of a $m \times n$ matrix A is $im(A) = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\}$

Theorem

If A is an $m \times n$ matrix, then $rank(A) + nullity(A) = n$.

Theorem

$$im(A) = col(A)$$

and then

$$dim(im(A)) = rank(A)$$

In consequence $dim(null(A)) + dim(im(A)) = n$

Example

Find a basis of the null space of a matrix $A = \begin{bmatrix} 1 & -2 & 1 & 1 \\ -1 & 2 & 0 & 1 \\ 2 & -4 & 1 & 0 \end{bmatrix}$ is to find a basis of the solution space of the homogeneous system $A\mathbf{x} = 0$.

The reduction of augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 2 & -4 & 1 & 0 & 0 \end{array}\right] \xrightarrow[r_2+r_1]{r_3-2r_1} \left[\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & -1 & -2 & 0 \end{array}\right] \xrightarrow{r_3+r_2} \left[\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The leading variables are x_1 and x_3, so the nonleading variables become parameters: $x_2 = s$ and $x_4 = t$. The general solution is

$$\mathbf{x} = \begin{bmatrix} 2s+t \\ s \\ -2t \\ t \end{bmatrix} = s\mathbf{x}_1 + t\mathbf{x}_2 \text{ where } \mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

So $\{\mathbf{x}_1, \mathbf{x}_2\}$ is a basis of $null(A)$ and then $dim(null(A)) = 2$

Find a basis of $im(A)$ is to find a basis of column space of A

In the row echolon matrix, the 1s leading are in columns 1 and 3. Hence the basis of $im(A)$ is $\left\{ \mathbf{c}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \mathbf{c}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

Exercise

Find a basis of null space, image space and rank of matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

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Coordinate

Let V be a n-dimension vector space and $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an **ordered basis** of V. For any vector $\mathbf{u} \in V$, there exists uniques tuple of numbers $(t_1, \ldots, t_n)$ such that

$$\mathbf{u} = t_1\mathbf{v}_1 + \cdots + t_n\mathbf{v}_n$$

We refers

$$[\mathbf{v}]_S = \begin{bmatrix} t_1 \\ \vdots \\ t_n \end{bmatrix}$$

as the **coordinate vector** of $\mathbf{v}$ with respect to the ordered basis S. The entries of $[\mathbf{v}]_S$ are called the coordinates of $\mathbf{v}$ with respect to S.

Example

Consider the vector $\mathbf{v} = 5x - 2$ in $\mathbf{P}_1$

1. In the order basis $S_1 = \{1, x\}$ of $\mathbf{P}_1$, we have

$$\mathbf{v} = -2 + 5x \Rightarrow [\mathbf{v}]_{S_1} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

2. In the order basis set $S_1 = \{x + 1, x - 1\}$, we have

$$\mathbf{v} = \frac{3}{2}(x + 1) + \frac{7}{2}(x - 1) \Rightarrow [\mathbf{v}]_{S_2} = \begin{bmatrix} \frac{3}{2} \\ \frac{7}{2} \end{bmatrix}$$

Exercise

Consider an ordered basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Find $[\mathbf{v}]_S$ if $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}$

Change Bases

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ be two ordered bases of a n-dimensional vector space V.

What is the connection between $[\mathbf{v}]_S$ and $[\mathbf{v}]_T$ for any vector v in the vector space V?

Change Bases

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ be two ordered bases of a n-dimensional vector space V.

What is the connection between $[\mathbf{v}]_S$ and $[\mathbf{v}]_T$ for any vector v in the vector space V? Suppose that

$$\mathbf{v} = c_1\mathbf{w}_1 + \cdots + c_n\mathbf{w}_n$$

that is $[\mathbf{v}]_T = \begin{bmatrix} c_1 & \ldots & c_n \end{bmatrix}^T$. Let

$$[\mathbf{w}_j]_S = \begin{bmatrix} a_{1j} & \ldots a_{nj} \end{bmatrix}^T$$

be the coordinate of $\mathbf{w}_j$ in S.

We have

$$
\begin{aligned}
\mathbf{v} \quad = \quad & c_1\left(a_{11}\mathbf{v}_1 + a_{21}\mathbf{v}_1 + \cdots + a_{n1}\mathbf{v}_n\right) \\
& + c_2\left(a_{12}\mathbf{v}_1 + a_{22}\mathbf{v}_2 + \cdots + a_{n2}\mathbf{v}_n\right) \\
& + \ldots \\
& + c_n\left(a_{1n}\mathbf{v}_1 + a_{n2}\mathbf{v}_2 + \cdots + a_{nn}\mathbf{v}_n\right) \\
& = (a_{11}c_1 + a_{12}c_2 + \cdots + a_{1n}c_n)\mathbf{v}_1 \\
& + (a_{21}c_1 + a_{22}c_2 + \cdots + a_{2n}c_n)\mathbf{v}_2 \\
& + \ldots \\
& \quad (a_{n1}c_1 + a_{n2}c_2 + \cdots + a_{nn}c_n)\mathbf{v}_n
\end{aligned}
$$

$$
\Rightarrow [\mathbf{v}]_S = \begin{bmatrix} a_{11}c_1 + a_{12}c_2 + \cdots + a_{1n}c_n \\ a_{21}c_1 + a_{22}c_2 + \cdots + a_{2n}c_n \\ \vdots \\ a_{n1}c_1 + a_{n2}c_2 + \cdots + a_{nn}c_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}
$$

Transition Matrix

$$P_{S\leftarrow T} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} [\mathbf{w}_1]_S & [\mathbf{w}_2]_S & \dots & [\mathbf{w}_n]_S \end{bmatrix}$$

is called the **transition matrix from the T-basis to the S-basis**

Change Base Coordinate

$$[\mathbf{v}]_S = P_{S\leftarrow T}[\mathbf{v}]_T$$

Example

Let V be R^3 and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ be ordered bases for R^3, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and

$$\mathbf{w}_1 = \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} 5 \\ 5 \\ 2 \end{bmatrix}.$$

(a) Compute the transition matrix $P_{S \leftarrow T}$ from the T-basis to the S-basis.

(b) Verify Equation (3) for $\mathbf{v} = \begin{bmatrix} 4 \\ -9 \\ 5 \end{bmatrix}$.

Solution

(a) First we need to find the $[\mathbf{w}_1]_S$, $[\mathbf{w}_2]_S$ and $[\mathbf{w}_3]_S$ by solving, for example

$$\mathbf{w}_1 = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3$$

We obtain

$$[\mathbf{w}_1]_S = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \quad [\mathbf{w}_2]_S = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad [\mathbf{w}_3]_S = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

So

$$P_{S\leftarrow T} = \begin{bmatrix} [\mathbf{w}_1]_S & [\mathbf{w}_2]_S & [\mathbf{w}_3]_S \end{bmatrix} = \begin{bmatrix} 2 & 2 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

(b) Compute directly we have

$$\mathbf{v} = 1\mathbf{w}_1 + 2\mathbf{w}_2 - 2\mathbf{w}_2 \Rightarrow [\mathbf{v}]_T = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

So

$$[\mathbf{v}]_S = P_{S \leftarrow T}[\mathbf{v}]_T = \begin{bmatrix} 2 & 2 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix}$$

Compute directly we obtain

$$[\mathbf{v}]_S = 4\mathbf{v}_1 - 5\mathbf{v}_2 + \mathbf{v}_3$$

which matches with the result obtained with the transition matrix.

Exercise

Consider $S = \{\mathbf{v}_1 = x, \mathbf{v}_2 = 1 - x\}]$ and $T = \{\mathbf{w}_1 = x - 1, \mathbf{w}_2 = x + 1\}$ be two bases in P_1. Find the transition matrix

$$P_{S \leftarrow T}$$

Property of Transition Matrices

Suppose that

$$P_{S\leftarrow T}[\mathbf{v}]_T = [\mathbf{0}]_S$$

If $[\mathbf{v}]_S = \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix}^T$ then

$$b_1\mathbf{v}_1 + \cdots + \mathbf{v}_n = \mathbf{0} \Rightarrow b_1 = \cdots = b_n = 0 \Rightarrow \mathbf{v} = \mathbf{0}$$

So $P_{S\leftarrow T}$ is invertible. Hence

$$[\mathbf{v}]_S = P_{S\leftarrow T}[\mathbf{v}]_T \Leftrightarrow [\mathbf{v}]_T = P_{S\leftarrow T}^{-1}[\mathbf{v}]_S$$

That is

$$P_{T\leftarrow S} = P_{S\leftarrow T}^{-1}$$

Transition Matrix from a Basis to Natural Basis

Consider $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ be two bases in $\mathbb{R}^n$. Let $E = \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the natural basis in $\mathbb{R}^n$. Denote

$$M_S = P_{E \leftarrow S} = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix}, \qquad M_T = P_{E \leftarrow T} = \begin{bmatrix} \mathbf{w}_1 & \ldots & \mathbf{w}_n \end{bmatrix}$$

For any vector $\mathbf{v} \in \mathbb{R}^n$, we have

$$[\mathbf{v}]_S = M_S^{-1}[\mathbf{v}]_E, \qquad \text{and} \qquad [\mathbf{v}]_E = M_T[\mathbf{v}]_E$$

So

$$[\mathbf{v}]_S = M_S^{-1}\left(M_T[\mathbf{v}]_T\right) = (M_S^1 M_T)\mathbf{v}_T$$

That is

$$P_{S \leftarrow T} = M_S^{-1} M_T$$

Example

Come back to the previous example. It is easy to see that

$$[\mathbf{v}_1]_E = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, [\mathbf{v}_2]_E = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, [\mathbf{v}_3]_E = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow M_S = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Similarly, we have

$$M_T = \begin{bmatrix} 6 & 4 & 5 \\ 3 & -1 & 5 \\ 3 & 3 & 2 \end{bmatrix}$$

So

$$P_{S \leftarrow T} = M_S^{-1} M_T = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 6 & 4 & 5 \\ 3 & -1 & 5 \\ 3 & 3 & 2 \end{bmatrix} = \dots$$